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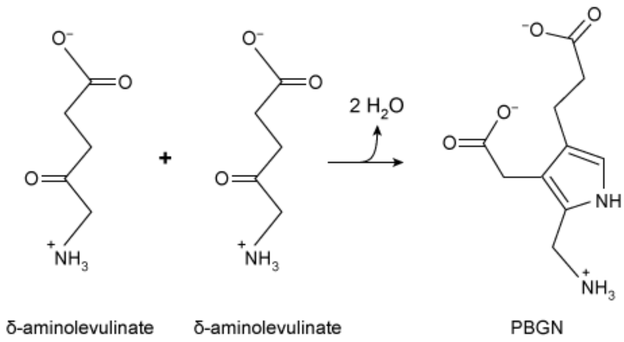
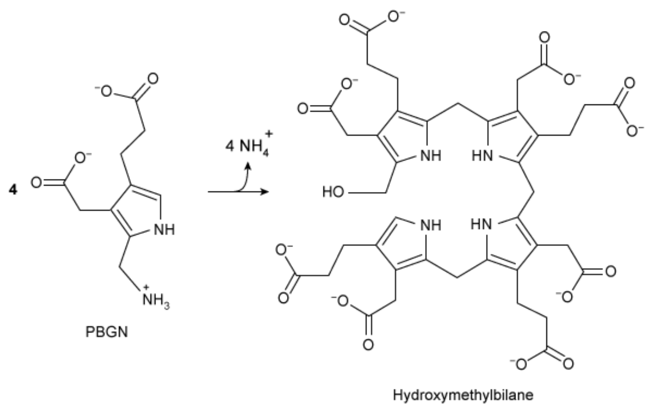


Reaction 1

In the next step, the monomeric enzyme hydroxymethylbilane synthase (HMBS) combines four PBGN molecules to form hydroxymethylbilane (Reaction 2). Several additional steps are required to convert hydroxymethylbilane to heme.



Reaction 2

Defects in HMBS activity can lead to accumulation of δ-aminolevulinate and porphobilinogen in the liver and blood plasma. The accumulation of these molecules can cause acute intermittent porphyria, a disorder characterized by episodes of severe abdominal pain. Researchers expressed, purified, and assessed several variants of HMBS for enzymatic activity (Table 1) and thermal stability (Figure 1).

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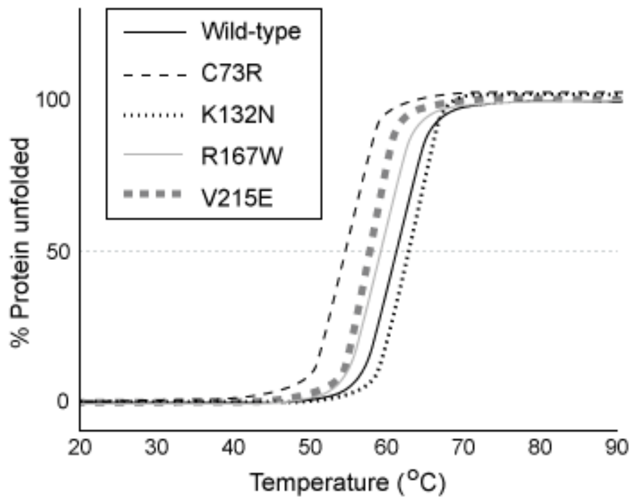


Figure 1 Thermal stability of several HMBS variants

What type of reaction does porphobilinogen synthase catalyze?

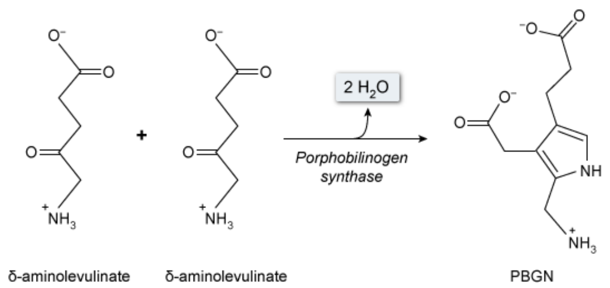
- A. Hydrolysis [9%]
- ✔ B. Condensation [85%]
- C. Deamination [4%]
- D. Methylation

Correct
86% answered correctly

微信Mai_ange

Explanation:

Condensation reactions join two molecules & release H_2O



Enzymes are biological catalysts that increase the rate of a reaction. They are broadly divided into six classes based on the general reaction types they catalyze. Each class is further divided into subclasses based on specific reaction types.

Lyases break bonds by forming double bonds or rings (double-bond equivalents) elsewhere. Porphobilinogen synthase is a lyase that combines two δ-aminolevulinate molecules by breaking C=O bonds and forming C=C bonds and an aromatic ring. Reaction 1 shows that two water molecules are released during bond formation in PBGN synthesis. Reactions in which molecules release water during bond formation are known as condensation reactions. Therefore, **porphobilinogen synthase** catalyzes a **condensation reaction**.

(Choice A) Hydrolysis reactions *consume* water to break bonds. Reaction 1 shows that porphobilinogen synthase catalyzes a reaction that *produces* water (a condensation reaction).

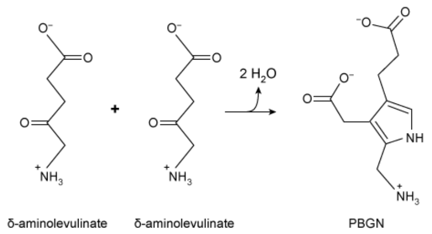
(Choice C) Deamination is the removal of one or more amino groups from a molecule, typically in the form of ammonia (NH_4^+). Deamination occurs in Reaction 2, catalyzed by HMBS, but does *not* occur in the reaction catalyzed by porphobilinogen synthase.

(Choice D) Methylation is the addition of a methyl group ($-CH_3$) to a molecule. None of the reactions shown in the passage are methylation reactions.

Educational objective:

Enzymes are biological catalysts that are classified according to the reactions they catalyze. Condensation reactions join two molecules together, and release one or more water molecules in the process.

Porphyrin biosynthesis is a multistep process that is required for production of porphyrin derivatives such as heme, which is a crucial component of hemoglobin, myoglobin, and the cytochrome cofactors of the electron transport chain. In the first step of this pathway, the enzyme porphobilinogen synthase catalyzes the production of porphobilinogen (PBGN) from two δ-aminolevulinate molecules, as shown in Reaction 1.



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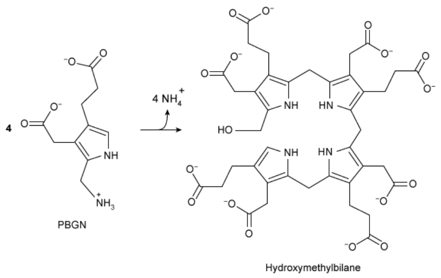

Reaction 2

Defects in HMBS activity can lead to accumulation of δ-aminolevulinate and porphobilinogen in the liver and blood plasma. The accumulation of these molecules can cause acute intermittent porphyria, a disorder characterized by episodes of severe abdominal pain. Researchers expressed, purified, and assessed several variants of HMBS for enzymatic activity (Table 1) and thermal stability (Figure 1).

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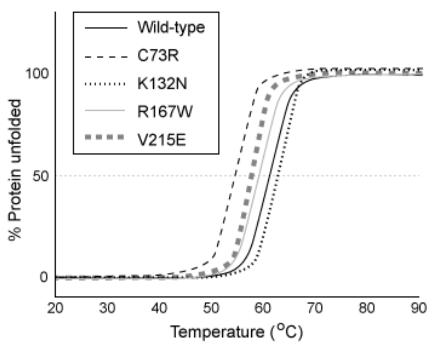
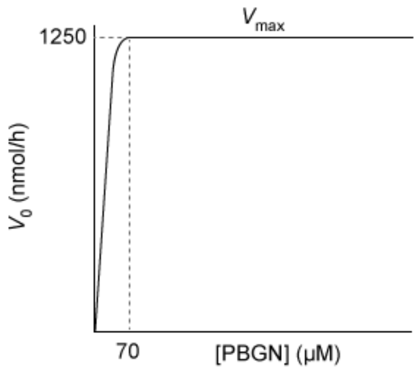


Figure 1 Thermal stability of several HMBS variants

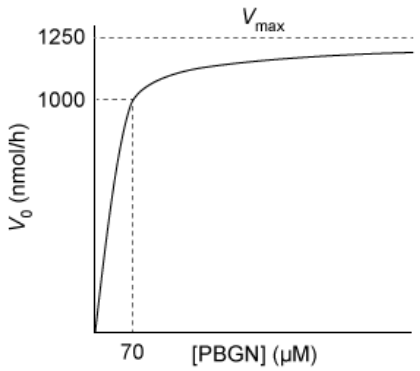
Which Michaelis-Menten plot represents the kinetics of wild-type HMBS?

A.



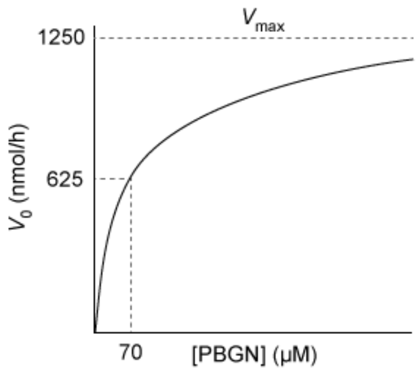
[6%]

B.



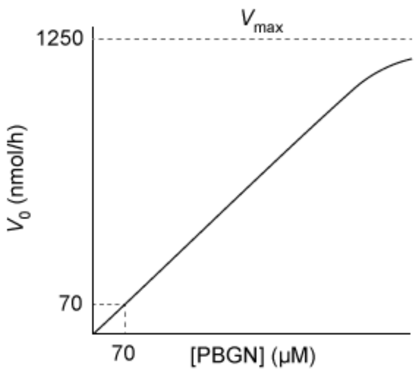
[6%]

✔ C.



[85%]

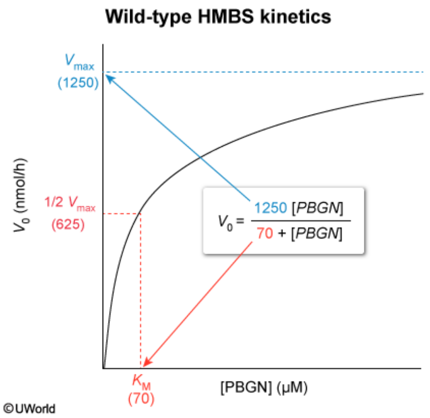
D.



[1%]

Correct
85% answered correctly

Explanation:



The **Michaelis-Menten equation** describes the rate of an enzymatic reaction as a **function of substrate concentration**. It is written in the form

$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

where [S] is the substrate concentration, V_0 is the observed reaction rate at a given [S], V_{max} is the maximum possible reaction rate, and K_m is the Michaelis constant. As [S] increases, V_0 increases hyperbolically, approaching a **plateau at V_{max}**. Table 1 shows that wild-type HMBS has a V_{max} of 1250 nmol/h. Accordingly, the graph should approach a plateau at 1250 nmol/h.

K_m and [S] are both measured in terms of concentration (eg, µM). **When [S] is equal to K_m**, the Michaelis-Menten equation can be rewritten as

$$V_0 = \frac{V_{max} K_m}{K_m + K_m} =$$

The K_m terms cancel, leaving

$$V_0 = \frac{V_{max}}{2}$$

Therefore, when [S] is equal to K_m, the **reaction proceeds at ½ V_{max}**. Table 1 shows that the K_m of wild-type HMBS is 70 µM, so **when [S] is 70 µM**, the reaction velocity is ½ the maximum velocity. In the case of wild-type HMBS, V_{max} is 1250 nmol/h, so **½ V_{max} = 1250/2 = 625 nmol/h**. Choice C shows a Michaelis-Menten graph with the correct V_{max} and K_m.

(Choice A) This graph shows that when [S] = 70 µM, the reaction proceeds at 1250 nmol/h, which is equal to V_{max}. In this scenario, ½ V_{max} occurs below 70 µM PBGN, indicating a K_m that is *lower* than 70 µM.

(Choice B) This graph shows that when [S] = 70 µM, the reaction proceeds at 1000 nmol/h, which is greater than ½ V_{max}. In this scenario, ½ V_{max} occurs below 70 µM PBGN, indicating a K_m that is *lower* than 70 µM.

(Choice D) This graph shows that when [S] = 70 µM, the reaction proceeds at 70 nmol/h, well below ½ V_{max}. In this scenario, ½ V_{max} occurs above 70 µM PBGN, indicating a K_m that is *higher* than 70 µM.

Educational objective:
The Michaelis-Menten equation describes the rate of an enzymatic reaction as a function of substrate concentration. The maximum reaction velocity V_{max}, found in the numerator of the equation, indicates the height of the plateau that the graph approaches. The K_m, found in the denominator, is the substrate concentration at which ½ V_{max} is achieved.

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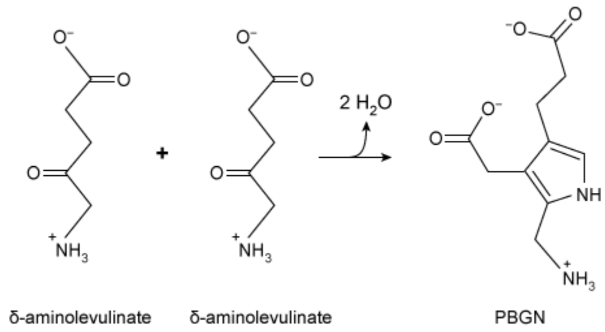

Reaction 1

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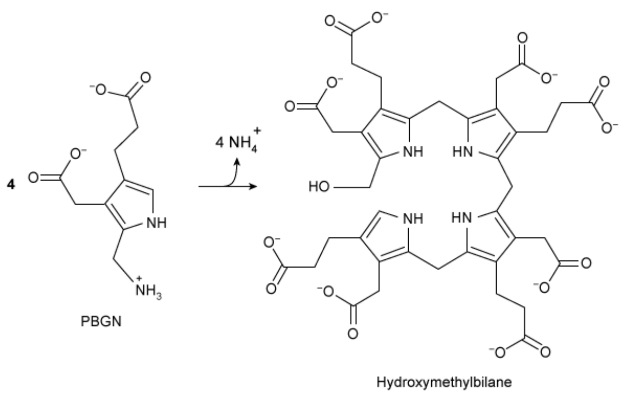

Reaction 2

Defects in HMBS activity can lead to accumulation of δ-aminolevulinate and porphobilinogen in the liver and blood plasma. The accumulation of these molecules can cause acute intermittent porphyria, a disorder characterized by episodes of severe abdominal pain. Researchers expressed, purified, and assessed several variants of HMBS for enzymatic activity (Table 1) and thermal stability (Figure 1).

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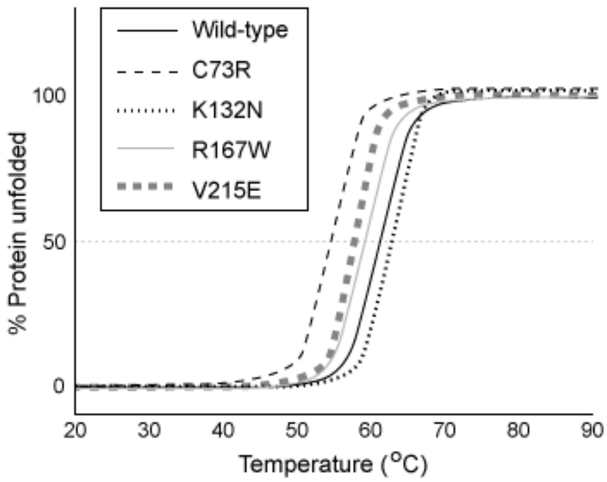

Figure 1 Thermal stability of several HMBS variants

Which statement most accurately describes the binding of PBGN to HMBS?
(Note: Assume binding is significantly faster than catalysis.)

- A. PBGN has a higher affinity for the wild-type enzyme than for any other enzyme form. [5%]
- B. PBGN has a higher affinity for the K132N variant than for the V215E variant. [19%]

- [x] C. PBGN has a lower affinity for the R167W variant than for any other enzyme form. [70%]
- [ ] D. PBGN binds all forms of the enzyme with approximately equal affinity. [4%]

Correct
70% answered correctly

Explanation:

K_M is an approximate measure of binding affinity

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R167W	$V_0 = \dfrac{680[PBGN]}{\underbrace{1600}_{K_M} + [PBGN]}$ Highest Km; lowest binding affinity
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Michaelis-Menten kinetics rely on a reaction model in which free enzyme E and free substrate S reversibly combine to form the enzyme-substrate complex ES. The rate of ES formation is proportional to the rate constant k_1, and the rate of dissociation is proportional to k_{-1}. After the ES complex forms, the enzyme catalyzes the formation of the product P. The rate of product formation is proportional to k_{cat}. **The Michaelis constant K_m** is defined as a combination of the rate constants:

$$K_m = \frac{k_{-1} + k_{cat}}{k_1}$$

When **binding** is **significantly faster** than **catalysis** k_{cat} is negligible, so **K_m can be approximated** as

$$K_m \approx \frac{k_{-1}}{k_1}$$

This expression is equal to the dissociation constant K_d, which measures binding affinity. Therefore, when binding is faster than catalysis, **K_m** is an **approximate measure of binding affinity**. Specifically, binding affinity *decreases* with *increasing* K_m.

Table 1 shows the Michaelis-Menten equations for each HMBS variant, with PBGN as the substrate. The number in the denominator of each equation is the K_m for that variant, measured in µM. The K_m of the R167W variant is 1600 µM, which is substantially *higher* than the K_m of any other variant tested. Therefore, PBGN has a **lower affinity** for the **R167W variant** than for any other form of the HMBS enzyme tested.

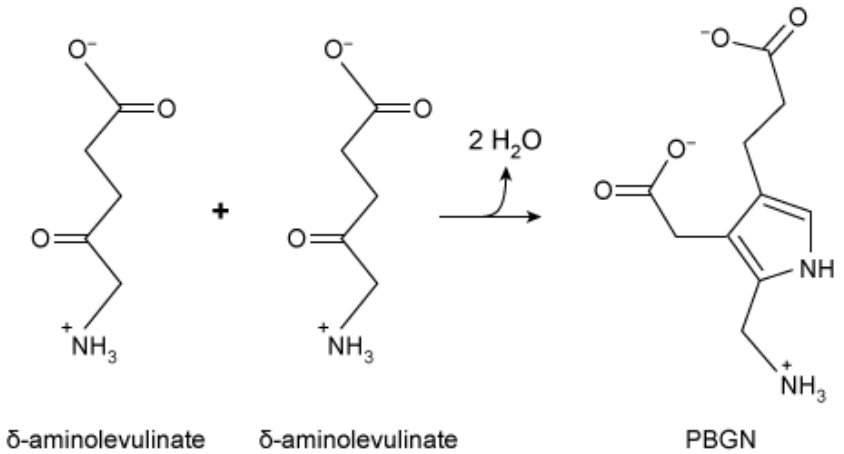
(Choice A) The K_m of PBGN for the wild-type enzyme is 70 µM. Both K132N and V215E have *lower* K_m values, and therefore PBGN has a *higher* affinity for these variants than for wild-type.

(Choice B) The K132N variant has a *higher* K_m value (63 µM) than the V215E variant (59 µM), so PBGN has a *higher* affinity for the V215E variant.

(Choice D) The K_m values listed range from 59 µM to 1600 µM, so the affinity of PBGN for the enzyme varies significantly between different forms of the enzyme.

Educational objective:
For enzymatic reactions in which substrate binding is significantly faster than catalysis, the Michaelis constant K_m is an approximate measure of binding affinity. A low K_m corresponds to high affinity and a high K_m corresponds to low affinity. The K_m value is found in the denominator of the Michaelis-Menten equation.

Porphyrin biosynthesis is a multistep process that is required for production of porphyrin derivatives such as heme, which is a crucial component of hemoglobin, myoglobin, and the cytochrome cofactors of the electron transport chain. In the first step of this pathway, the enzyme porphobilinogen synthase catalyzes the production of porphobilinogen (PBGN) from two δ-aminolevulinate molecules, as shown in Reaction 1.



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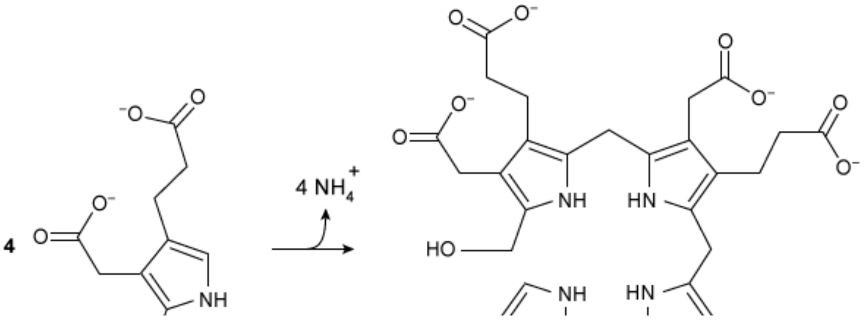

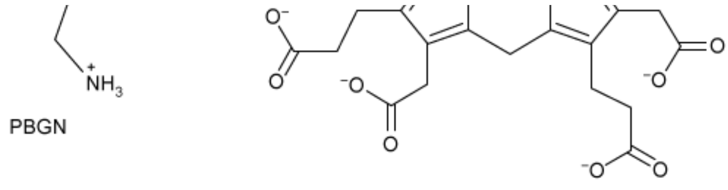

PBGN

Hydroxymethylbilane

Reaction 2

Defects in HMBS activity can lead to accumulation of δ-aminolevulinate and porphobilinogen in the liver and blood plasma. The accumulation of these molecules can cause acute intermittent porphyria, a disorder characterized by episodes of severe abdominal pain. Researchers expressed, purified, and assessed several variants of HMBS for enzymatic activity (Table 1) and thermal stability (Figure 1).

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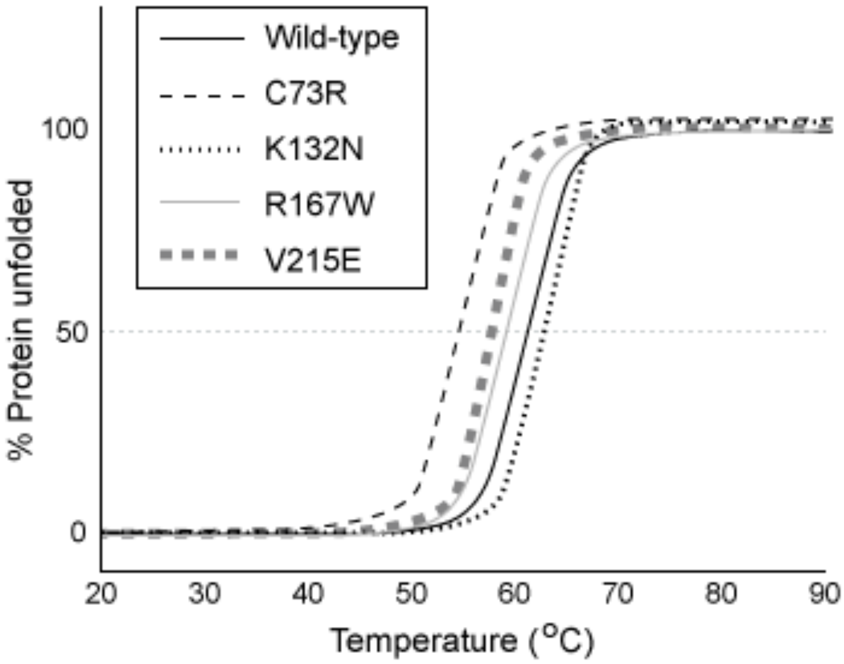

Figure 1 Thermal stability of several HMBS variants

Which HMBS mutant variant is LEAST likely to cause acute intermittent porphyria?

- A. C73R [6%]
- B. K132N [58%]
- C. R167W [11%]
- D. V215E [23%]

Correct
58% answered correctly

Explanation:

Comparison of wild-type & mutant HMBS enzyme activities

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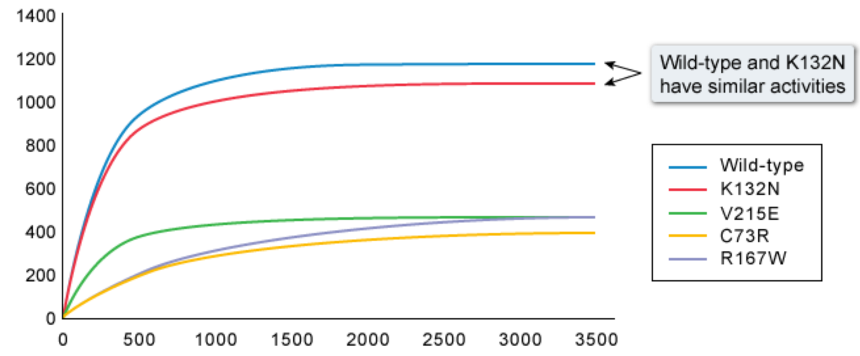
Similar V_{max} and K_M values



Genetic disorders commonly occur when one or more **mutations in a gene** cause the **protein** expressed by that gene to have **impaired function**. Different genetic mutations may impair function to varying degrees. Some mutations completely inactivate the protein, whereas others essentially have no effect. To **determine how likely a mutation** is to **cause a genetic disorder**, the **activity** of the mutant protein should be **compared to the wild-type form**.

The passage states that defects in HMBS can lead to accumulation of δ-aminolevulinate and porphobilinogen, which can cause acute intermittent porphyria. Accumulation most likely occurs because HMBS fails to catalyze conversion of PBGN to hydroxymethylbilane. The Michaelis-Menten equations in Table 1 indicate how quickly each HMBS variant catalyzes the reaction.

The numerator of each equation shows the maximum possible reaction velocity (V_{max}) for each variant. According to the table, the V_{max} of the K132N variant (1160 nmol/h) is the closest to the V_{max} of the wild-type form (1250 nmol/h). The K_m of the K132N variant (found in the denominator) is also similar to the K_m of the wild-type form. Therefore, of the mutant variants shown, **K132N** exhibits **activity** that is **most similar to wild-type HMBS** and would be **least likely** to cause acute intermittent porphyria.

(Choices A, C, and D) Table 1 shows that the C73R, R167W, and V215E mutants each have significantly lower V_{max} values than wild-type HMBS, and therefore they catalyze the reaction much more slowly. A decreased reaction rate would lead to accumulation of reactants (ie, PBGN), so these variants are *likely* to cause acute intermittent porphyria.

Educational objective:
Genetic disorders are commonly caused by mutations to genes that reduce the activity of the proteins they express. The impact of a mutation can be assessed by comparing the activities of mutant forms of the protein to the activity of the wild-type form.

Subject:
Biochemistry

System:
Enzymes

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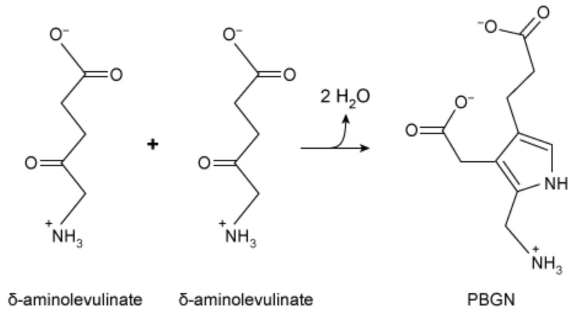

Reaction 1

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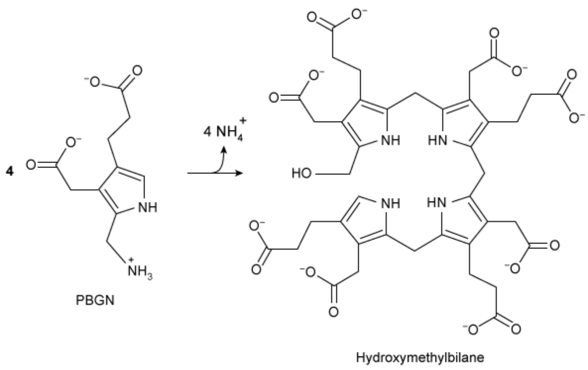

Reaction 2

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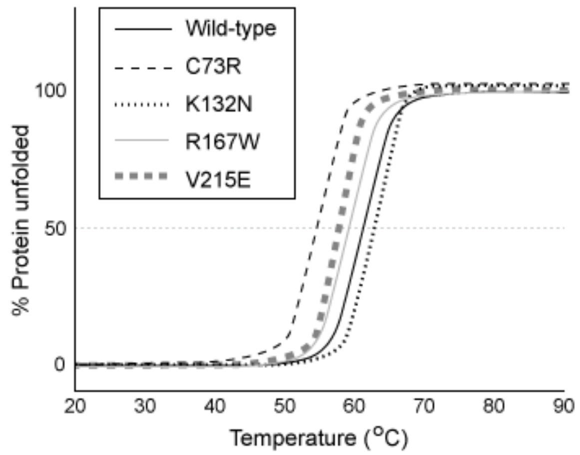


Figure 1 Thermal stability of several HMBS variants

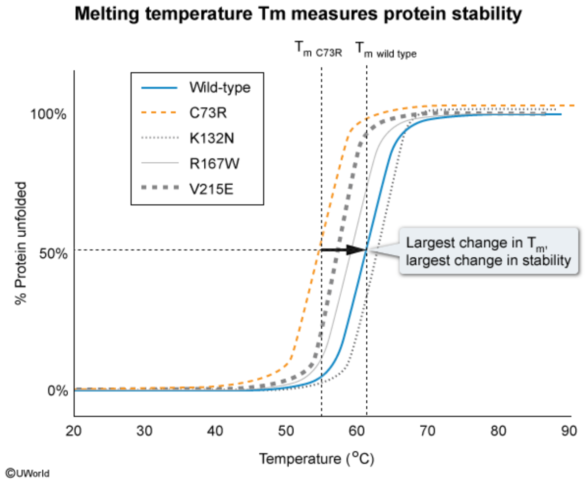
Based on the data in Figure 1, a mutation of which amino acid residue has the greatest impact on HMBS stability?

- A. Cysteine at position 73 [71%]
- B. Lysine at position 132 [10%]
- C. Arginine at position 167 [12%]
- D. Valine at position 215 [4%]

Correct
70% answered correctly

Explanation:

微信Mai_ange



The **stability of a protein** can be measured by its **ability to remain folded** as ambient temperature increases. An increase in temperature causes the atoms within a protein to move faster and increases the probability that **intramolecular interactions** such as hydrogen bonds, ionic attractions, and other stabilizing forces will be disrupted, causing the protein to unfold (denature). Proteins that **denature at higher temperatures** have stronger intramolecular forces and are **more stable**.

The **temperature at which 50%** of the proteins in solution **become denatured** is known as the melting temperature (T_m). A **higher T_m** indicates a **more stable protein**. Figure 1 shows the T_m of several different HMBS variants. The wild-type form of the enzyme has a T_m slightly above 60 °C. To determine which mutation has the greatest impact on HMBS stability (T_m), the T_m of each HMBS variant must be compared to the wild-type T_m. The T_m of the C73R variant is approximately 55 °C and is further away from the wild-type T_m than any other variant shown. The C73R designation indicates that the cysteine residue (C) at position 73 was mutated to arginine (R). Therefore, mutating the cysteine residue at position 73 has a greater impact on HMBS stability than any of the other mutations listed.

(Choices B, C, and D) The K132N, R167W, and V215E variants have T_m values that are closer to the wild-type value than that of the C73R variant. Therefore, mutating lysine at position 132, arginine at position 167, or valine at position 215 has a smaller impact on stability than mutating cysteine at position 73.

Educational objective:
Protein stability is determined by the strength of the intramolecular forces that maintain a protein in its folded form. The temperature at which 50% of proteins in solution become denatured (melting temperature T_m) is a measure of protein stability. A higher T_m corresponds to a more stable protein.